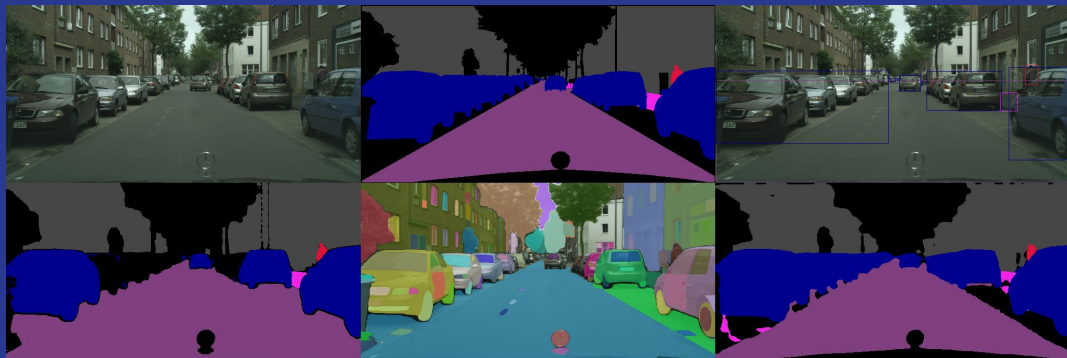
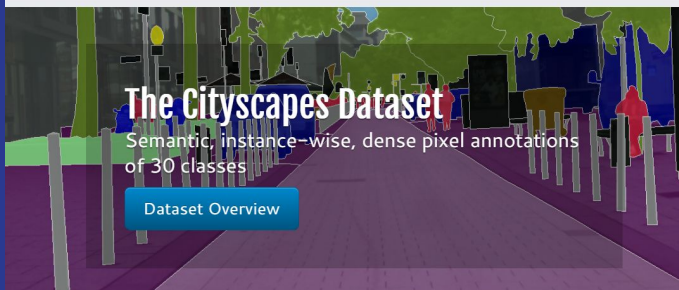


Segment Anything Model (SAM) for Cityscapes

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Segment Anything

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Figure 1: We aim to build a foundation model for segmentation by introducing three interconnected components: a promptable segmentation task, a segmentation model (SAM) that powers data annotation and enables zero-shot transfer to a range of tasks via prompt engineering, and a data engine for collecting SA-1B, our dataset of over 1 billion masks.

Abstract

We introduce the Segment Anything (SA) project: a new task, model, and dataset for image segmentation. Using our efficient model in a data collection loop, we built the largest segmentation dataset to date (by far), with over 1 billion masks on 11M licensed and privacy-respecting images. The model is designed and trained to be promptable, so it can transfer zero-shot to new image distributions and tasks. We evaluate its capabilities on numerous tasks and find that its zero-shot performance is impressive – often competitive with or even superior to prior fully supervised results. We are releasing the Segment Anything Model (SAM) and corresponding dataset (SA-1B) of 1B masks and 11M images at <https://segment-anything.com> to foster research into foundation models for computer vision.

Empirical trends show this behavior improving with model scale, dataset size, and total training compute [56, 10, 21, 93]. Foundation models have also been explored in computer vision, albeit to a lesser extent. Perhaps the most prominent illustration aligns paired text and images from the web. For example, CLIP [52] and ALIGN [53] use contrastive learning to train text and image encoders that align the two modalities. Once trained, engineered text prompts enable zero-shot generalization to novel visual concepts and data distributions. Such encoders also compose effectively with other models to enable downstream tasks, such as image generation (e.g., DALL-E [83]). While much progress has been made on vision and language encoders, computer vision includes a wide range of problems beyond this scope, and for many of these, abundant training data does not exist.

In this work, our goal is to build a foundation model for image segmentation. That is, we seek to develop a promptable model and pre-train it on a broad dataset using a task that enables powerful generalization. With this model, we aim to solve a range of downstream segmentation problems on new data distributions using prompt engineering.

The success of this plan hinges on three components: **task, model, and data**. To develop them, we address the following questions about image segmentation:

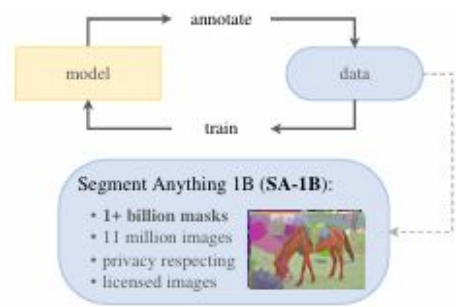
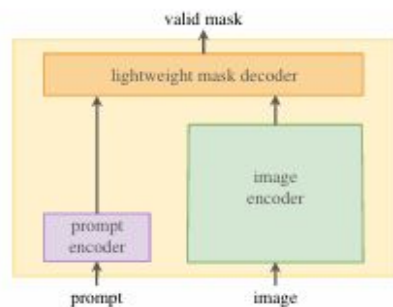
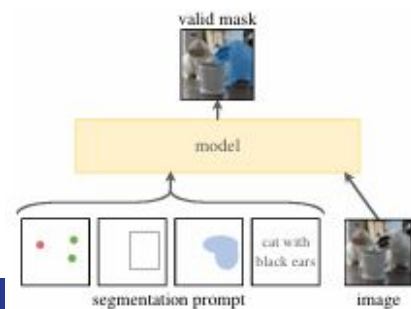
1. What task will enable zero-shot generalization?
2. What is the corresponding model architecture?
3. What data can power this task and model?

Motivation and Task

- Image segmentation underpins many tasks:
 - Scene understanding, autonomous driving, AR, image editing.
- Typical segmentation models:
 - Trained for one dataset or one task.
 - Require large, expensive pixel-level labels.
- Goal of Segment Anything (SAM):
 - Promptable segmentation that can generalize across tasks and datasets.
- Our focus:
 - Evaluate SAM on urban driving scenes from the Cityscapes dataset.
 - Study domain adaptation: can a Cityscapes-tuned SAM head beat a strong zero-shot baseline?

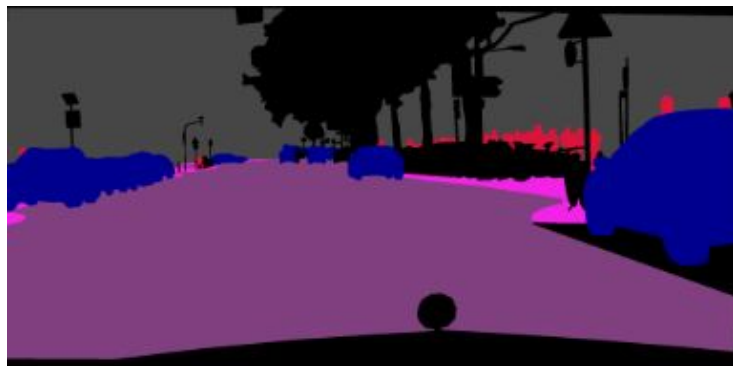
Segment Anything

- Novel task: Promptable Segmentation
 - Given an image and a prompt, SAM returns a valid segmentation mask for any image.
 - Prompts: points, bounding boxes, masks, text (in follow-up work)
- Segment Anything Model (SAM)
 - MAE-pretrained ViT-H image encoder.
 - Prompt encoder for points, boxes, masks.
 - Lightweight mask decoder that fuses image and prompt tokens
- SA-1B dataset
 - ~1.1B masks on 11M images
 - Model-in-the-loop annotation pipeline



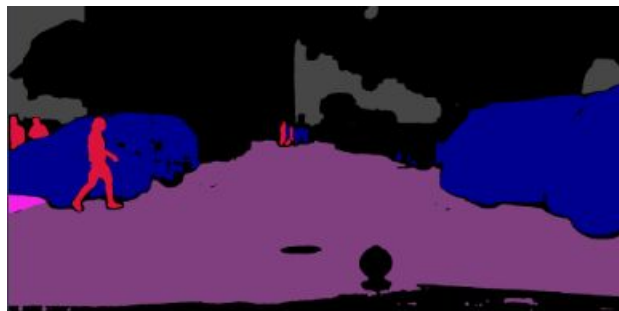
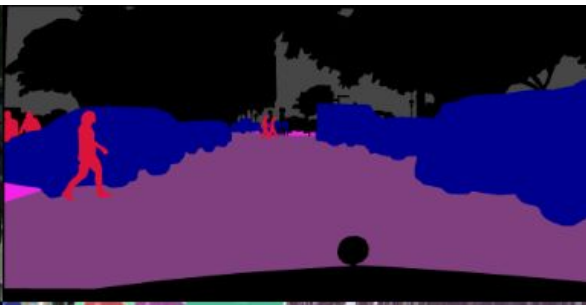
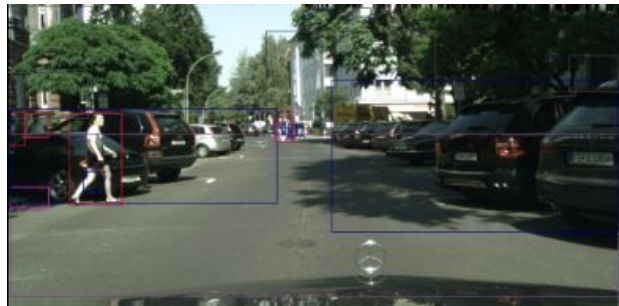
Our Downstream Task

- Dataset:
 - Cityscapes: 2,975 train and 500 val segmented images of urban street scenes into 19 classes
- Downstream task:
 - Segmentation of urban driving scenes
 - Focused on only 5 classes: road, sidewalk, car, person, building
 - Research question: "How does domain adaptation affect SAM's performance?"
 - Zero-shot SAM on Cityscapes vs. Cityscapes-tuned SAM with a new semantic head
- Evaluation metrics
 - Intersection-over-Union of each class
 - Mean Intersection-over-Union (mIOU)



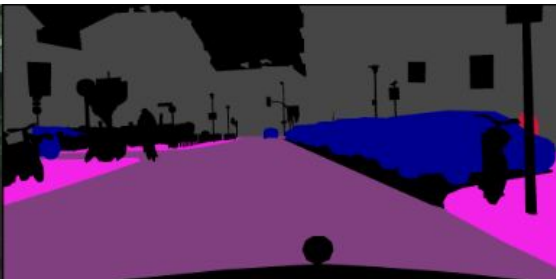
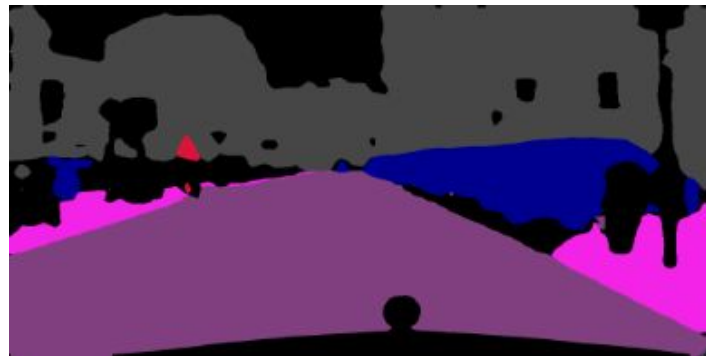
Zero-Shot SAM Baseline

- Box-Prompt Generation
 - Extract ground-truth masks from Cityscapes
 - Sample tight bounding boxes around class regions
- Using SAM
 - Use pretrained weights checkpoint from SAM's Github
 - Feed boxes to SAM's box prompt interface using SamPredictor
 - Interpret SAM's output and segment image
- Result: 0.7192 mIoU



Our Cityscapes-Tuned SAM Head

- Use SAM's image encoder as a frozen generic feature extractor
- Attach a lightweight semantic head that predicts 6 classes:
 - Road, sidewalk, building, person, car, or "none of the above"
 - Head architecture: Two 3x3 convolutional layers, batch normalization, ReLU layers
- Training
 - Loss: cross-entropy
 - Optimizer: Adam
 - Learning rate: 1e-3, Epochs: 5, Batch size: 2
 - Trained on all 2975 images, evaluated on all 500 images
 - RTX 3080 GPU, ~4 hours
- Result: 0.8013 mIoU



Quantitative Results

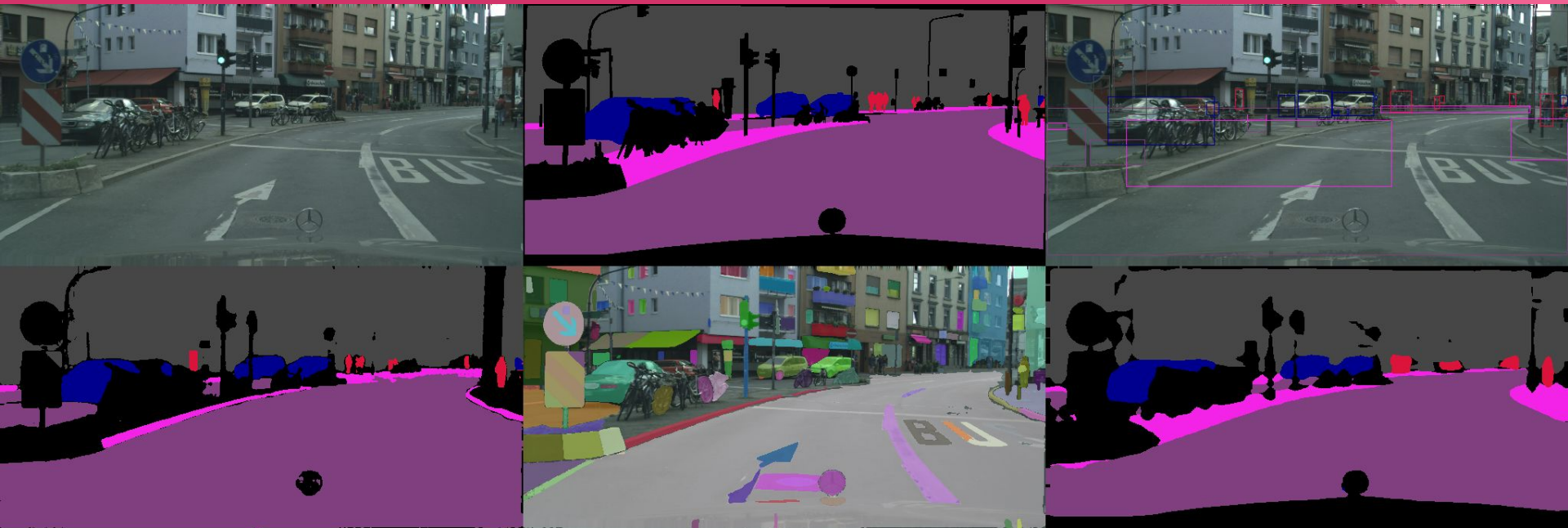
- Box-prompt zero-shot SAM baseline

- mIoU: 0.7192
- Per-class IoU
 - road: 0.8544
 - sidewalk: 0.6984
 - building: 0.6944
 - person: 0.6114
 - car: 0.7374

- SAM-based Cityscapes-tuned semantic head

- mIoU: 0.8013
- Per-class IoU
 - road: 0.9256
 - sidewalk: 0.6990
 - building: 0.8535
 - person: 0.6555
 - car: 0.8731

Qualitative Results



Analysis and Discussion

Why did the tuned head get better results?

- Tuned head was trained on the same dataset it was tested on
- SAM is only as good as its prompt and good prompt generation is not easy
- SAM is strong at single-object masks with high-quality prompts
- While the tuned head's output is not as sharp, it “understands” the task better than SAM and scores higher on IoU, so SAM domain adaptation is a worthwhile project for specific downstream tasks

However:

- SAM's zero-shot results are impressive and with better prompts it could perform better than our model